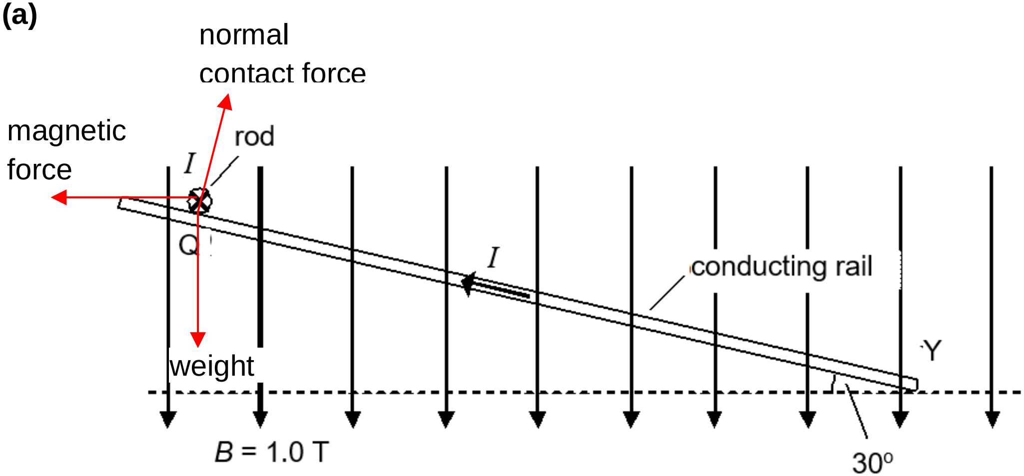
7 **(a)**



A1

(b) $F_B = BIL = (1)(0.50)(0.50) = 0.25$ N — C1

Considering forces along the slope,
$mg\sin 30 - F_B\cos 30 = ma$
$0.030(9.81)\sin 30 - (0.25)\cos 30 = 0.030a$ — C1
$a = -2.31$ m s^{-2} — A1

direction: parallel to the slope, away from Y — B1

(c) Initially, the rod will accelerate up the slope so its speed will increase. — B1

As its speed increases, there is cutting of magnetic field lines and resulted in induced emf in the rod according to Faraday's law. — B1

There will be induced current in the rod that flows in direction that generates magnetic field and force that opposes the motion of rod according to Lenz's Law. — B1

Hence, the net magnetic force acting on the rod up the slope will reduce and the rod will increase its speed until it reaches terminal velocity. — B1